Performing FFT and Inverse FFT.

```
%Defining Parameters
L = 2*pi; % sampling points on circle
N = 64;
dx = L/N;
x = (0:N-1)*dx;

%Periodic Function
u=exp(-sin(x/2).^2)
```

```
u = 1×64
    1.0000    0.9976    0.9904    0.9787    0.9627    0.9427    0.9192    0.8927 ···
```

```
%Fourier Transform
u_hat=fft(u);
```

Now, we want to decide a correct wavenumber, if not the spectral method will become nonsense

Logic:

1) Manually map the discrete indices to physical frequencies

2) Hence, we want to choose the smallest possible non-zero frequency(fundamental). Here is should be 2*pi/L.

3) On a grid of "N" points, the computer can only "see" a certain maximum speed (the **Nyquist frequency**). Any wave faster than that "aliases" and looks like a slower wave moving in the opposite direction. (take 1 to 8 index)

Index	N-Value	Name	Physical Meaning
1	0	"DC" or no wave Component	Average or constant offset of the function
2,3,4	1,2,3	Positive Modes	Waves that complete 1,2,3, full cycles in the domian L
5	0	Nyquist(Maximum)	The wave with maximum speed.
6,7,8	-3,-2,-1	Negative Modes	The mirrors of the positive modes-same frequencies but different phase/direction

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Why the Nyquist freq is forced as zero? Technically, it should be 4. But, k=4 is the excat limit of what the grid can solve. As in, for the middle, N/2 and -N/2 are the same, which is not the case for positive modes, i.e. for a positive mode, say 1, we have -1 which is a negative mode, and both are distinct. But, 4 and -4, in the circle are the same-the phase information is lost! The positive mode(say 1) and its conjugate(-1) when combined together, give a real entity. But, such is not the case for N/2. Hence, we force it to be 0, so that it doesnt become a complex exponential.

(Remark: When we differentiate, we will get an ik component. Hence, we get purely imaginary qty. while doing the inverse FT, it has no conjugate partner to cancel the imaginary part.)

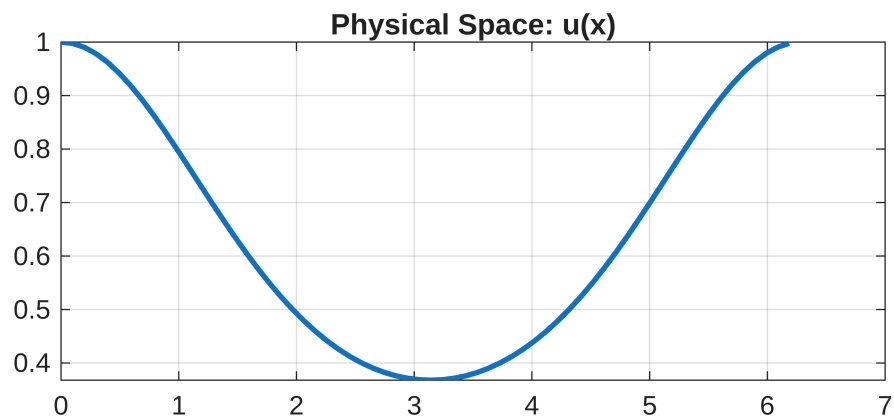
```
% 2. Define the correct Wavenumber vector (k)
% This matches the order of fft(u)
k = (2*pi/L) * [0:N/2-1, 0, -N/2+1:-1]';
% []' the quotation mark is for transposing

% 3. Inverse Transform to get back to physical space
u_recovered = real(ifft(u_hat));
```

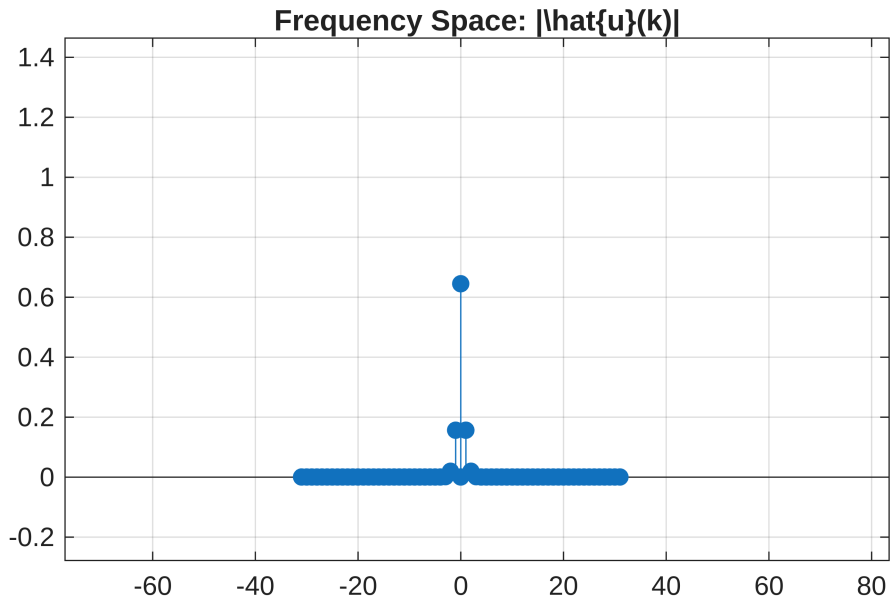
We obtained our matrices. Now, to visualize our result, we will plot the respective functions:

```
% --- VISUALIZATION ---
figure('Position', [100, 100, 1000, 400]);

% Plot 1: Physical Space (The "Function")
figure(1);
plot(x, u, 'LineWidth', 2);
title('Physical Space: u(x)');
grid on;
```




```
% Plot 2: Fourier Space (The "Matrix/Coefficients")
% We use fftshift to put 0 in the middle for easier viewing
figure(2);
stem(fftshift(k), abs(fftshift(u_hat))/N, 'filled');
title('Frequency Space: |\hat{u}(k)|', 'Interpreter', 'none');
grid on;
```


Frequency Space: |\hat{u}(k)|
1.4
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
-60
-40
-20
0
20
40
60
80